Journal of the Royal Statistical Society Series B, 2026, , 1–2
Supplementary material

Supplementary material for “A closed-form goodness-of-fit reference for penalised logistic regression in the proportional regime”

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Abstract

This document reports the material referred to in the paper: the behaviour of the correction under penalties other than the ridge, and under predictor laws other than the Gaussian.

Keywords calibration; goodness-of-fit; penalised regression

S1. Penalties other than the ridge

Nothing in Sections 2–4 of the paper uses the linearity of the ridge penalty. If a smooth penalty contributes a score term $\boldsymbol{a}(\boldsymbol{\beta})$ with $K(\boldsymbol{\beta}) = -\partial \boldsymbol{a}/\partial \boldsymbol{\beta}$, then $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - F^{-1}\boldsymbol{a}(\hat{\boldsymbol{\beta}})$, $\hat{\boldsymbol{\mu}} = -UM^{-1}\boldsymbol{a}(\tilde{\boldsymbol{\beta}})$ and Theorem 1 of the paper go through with K evaluated at $\hat{\boldsymbol{\beta}}$ (Ebrahim, 2026, Remark 2). We implemented this for the generalised ridge penalty $\sum_j \theta_j \beta_j^2$ and for Firth's penalty (Firth, 1993), whose score contribution is the Jeffreys term $a_j(\boldsymbol{\beta}) = \frac{1}{2}\,\mathrm{tr}(F^{-1}\partial F/\partial \beta_j)$ and which is the standard response to separation in clinical prediction (Puhr et al., 2017); the Firth fit agrees with the `logistf` package (Heinze and Schemper, 2002) to 2×10^{-8}.

Table S1 makes three points. Under the generalised ridge the picture is the ridge picture: the uncorrected test rejects almost every correct model, the first-order reference is conservative by the deflation of Theorem 1 of the paper (ratio 0.93 at fixed p and 0.83 at $\kappa = 0.25$), and the variance at the debiased fit restores the level at fixed p (0.060 and 0.047) and leaves at $\kappa = 0.25$ the same two-to-three-point excess (0.080 and 0.070) that Section 4 of the paper treats. Under Firth's penalty at fixed p there is nothing to correct: the Jeffreys term is $O(n^{-1})$, the displacement is negligible, and the three references coincide at 0.067 on the decile basis and 0.053 on the EDGE basis. At $\kappa = 0.25$ Firth's penalty is not a shrinkage penalty. The fitted coefficient norm is 3.51 against a true value of 2.31, the proportional-regime inflation of Sur and Candès (2019) that an $O(n^{-1})$ bias reduction does not touch, and the one-step debiasing moves further away from the truth (4.75), so the corrected references reject 12 to 32 per cent of correct models and the uncorrected test 99 per cent. The framework of this paper is for estimators that shrink towards zero. For bias-reduced estimators in the proportional regime the reference would have to be built on the inflation rather than on the displacement, and we leave that open.

S2. Predictor laws other than the Gaussian

Assumption 2 of the paper requires Gaussian predictors. The table below repeats three cells with t_3 and standardised Bernoulli(0.3) predictors, the correlation structure imposed through the same Cholesky factor, and reports the size of the de-noised reference and the estimated standard deviation of the linear predictor. Section 5.4 of the paper runs the whole procedure, rather than the reference alone, under the same two laws.

Table S1. Rejection rate of a correctly specified model at nominal level 0.05 under three smooth penalties, designs A ($\lambda = 100$) and B ($\kappa = 0.25$, $\lambda = 416$; the generalised ridge uses $\theta_j = \lambda(1 + j/p)$): the uncorrected statistic under the maximum likelihood covariance, the corrected statistic under the first-order reference Ω_{MLE}, and under the variance at the debiased fit Σ_{d}; last column, the deflation ratio $\mathrm{E}(\mathrm{SC.HL})/\,\mathrm{tr}(\Omega_{\mathrm{MLE}})$. 300 replications per cell.

		Decile basis			EDGE basis			
Design	penalty	uncorrected	first order	Σ_{d}	uncorrected	first order	Σ_{d}	deflation
A	ridge	0.937	0.027	0.043	1.000	0.023	0.043	0.91
A	generalised ridge	0.997	0.030	0.060	1.000	0.017	0.047	0.93
A	Firth	0.067	0.067	0.067	0.053	0.053	0.053	1.03
B, $\kappa = 0.25$	ridge	1.000	0.013	0.057	1.000	0.023	0.077	0.79
B, $\kappa = 0.25$	generalised ridge	1.000	0.023	0.080	1.000	0.027	0.070	0.83
B, $\kappa = 0.25$	Firth	0.987	0.117	0.317	1.000	0.127	0.297	1.15

Table S2. Size at nominal level 0.05 of the de-noised reference Σ_{dn} under three predictor laws, decile and EDGE-3 bases, with the estimated standard deviation of the linear predictor (true values 0.79 in design A and 1.48–1.50 in design B). 400 replications per cell.

Design	law	decile	EDGE-3	estimated s.d.
A	Gaussian	0.048	0.042	0.79
A	t_3	0.052	0.052	0.77
A	binary	0.058	0.060	0.80
B, $\kappa = 0.10$	Gaussian	0.060	0.045	1.45
B, $\kappa = 0.10$	t_3	0.062	0.098	1.38
B, $\kappa = 0.10$	binary	0.070	0.058	1.44
B, $\kappa = 0.25$	Gaussian	0.062	0.077	1.43
B, $\kappa = 0.25$	t_3	0.055	0.065	1.37
B, $\kappa = 0.25$	binary	0.080	0.052	1.43

S3. Nominal levels other than five per cent

Every size table in the paper reports the rejection rate of a correct model at the single nominal level 0.05, while Section 6 reads two p-values as magnitudes rather than as decisions at that level. Table S3 therefore repeats the check at 0.01 and 0.10, from the same replications; no new simulation is involved, and the per-replication p-values are in the reproduction archive.

Across the twenty-five cells the adaptive statistic rejects between 0.000 and 0.062 of correct models at the one per cent level and between 0.030 and 0.154 at the ten per cent level, so its validity is not an artefact of the level at which it was examined; it is conservative in the far tail, returning no rejection in 500 replications in three cells. The decile statistic behaves at 0.01 as it does at 0.05, including where it fails: in the design with covariate correlation 0.3 it rejects 0.390 of correct models at the one per cent level against 0.688 at five per cent, so that failure is a property of the reference and not of a threshold.

Table S3. Rejection rate of a correctly specified model at three nominal levels, for the decile statistic (CALM-D) and the adaptive EDGE statistic (CALM-E). Seven of the twenty-five cells of Tables 2 and 3 of the paper, including the design in which the decile statistic is known to fail. 500 replications per cell.

	CALM-D			CALM-E		
Design	0.01	0.05	0.10	0.01	0.05	0.10
A, $\kappa = 0.01$	0.008	0.046	0.096	0.016	0.050	0.072
B, $\kappa = 0.10$	0.008	0.060	0.110	0.002	0.040	0.094
B, $\kappa = 0.25$	0.006	0.052	0.102	0.000	0.028	0.060
B, $\kappa = 0.40$	0.010	0.036	0.088	0.004	0.018	0.048
B′, $\kappa = 0.33$	0.006	0.052	0.094	0.000	0.030	0.064
B′, $n = 800$, $\kappa = 0.25$	0.026	0.122	0.180	0.014	0.074	0.140
B′, correlation 0.3	0.390	0.688	0.798	0.062	0.076	0.102